Author: Sentao ZENG | Environmental Change and Global Sustainability | University of Milan | Jan 23, 2023

Dendrobates tinctorius

The Dyeing Poison Frogs and their place in the IUCN Red List

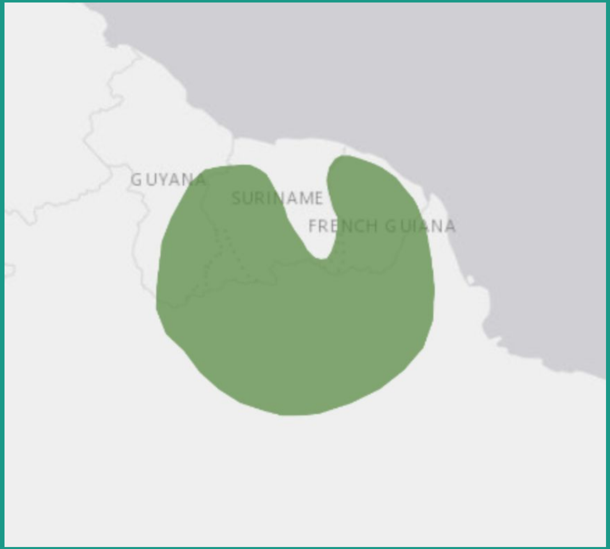



Species Overview

Among the largest species, the Dyeing Poison Frog is distributed throughout the eastern portion of the Guiana Shield, including parts of Guyana, Suriname, Brazil, and nearly all of French Guiana.

Habitat Type: Forest, Wetlands (inland)

It is illegally collected for the pet trade. Conservation protection over entire range.



Source: Map of Life (mol.org)

Assessing This Species

By using RStudio & IUCN Red List Criteria. Due to limited data source, we estimate AOO by the surface of suitable habitat.

Making <reclass_dentin.txt> to reflect the right habitat types & select column 1 & 4 only:

category	name	atefla	**dentin**	Pristimanti
1	urban	0	**0**	
13	orpaillage and mining activities	0	**0**	
21	arable	0	**0**	
22	permanent colture	0	**0**	
23	grassland	0	**0**	
24	heterogeneous colture	0	**0**	
31	forest	1	**1**	
32	savannah	0	**0**	
33	bare	0	**0**	
34	degraded forest	0	**1**	
41	inland wetland	0	**0**	
42	marine wetland	0	**0**	
51	inland water	0	**0**	
318	mangroves	0	**0**	
332	inselberg	0	**1**	

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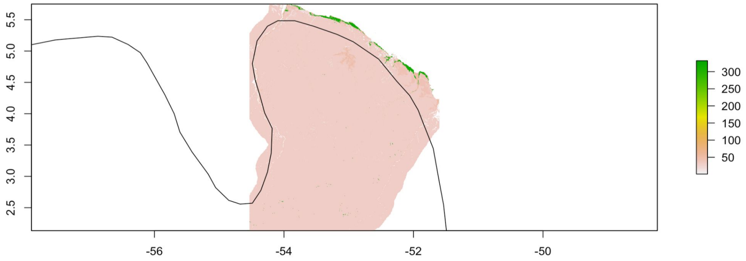
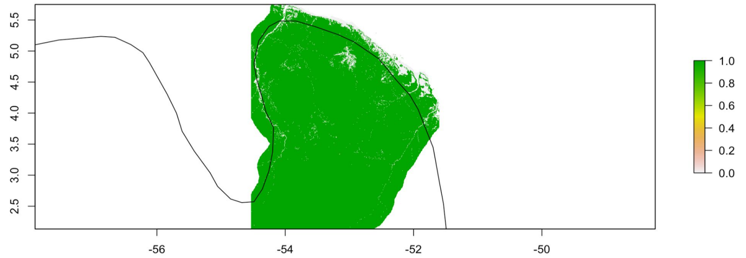
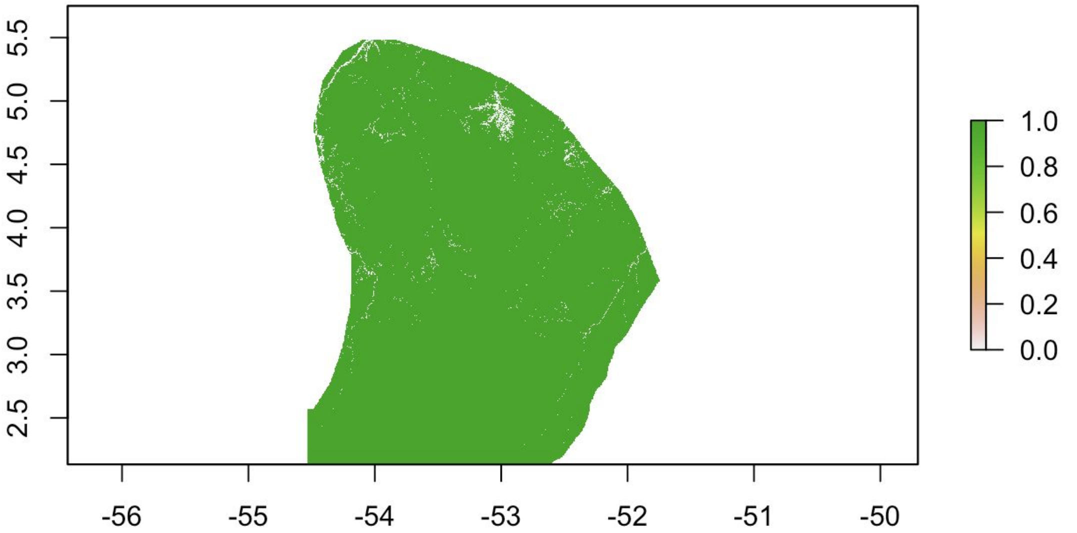
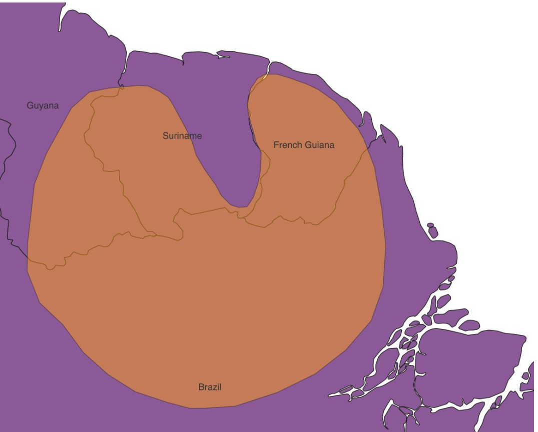








Suitable area of the species in 2012 (current) within French Guiana.

We use the land use data from 2005 as “past” and 2012’s as “current”.

Suitable Habitat in 2005 vs 2012

```
> ### calculate the suitable habitat in 2005
> a=area(suit_2005_dentin)
> zonal(a, suit_2005_dentin, "sum") ## calcula
     zone       sum
[1,]    0   1008.78
[2,]    1  83590.61
> 83590.61-83489.825
[1] 100.785
> 100.785*100/83590.61
[1] 0.1205698
```

```
> ### calculate the suitable habitat in 2012
> a=area(suit_2012_dentin)
> zonal(a, suit_2012_dentin, "sum") ## calculat
nd favorable (1) habitat
     zone        sum
[1,]    0   1111.487
[2,]    1  83489.825
```

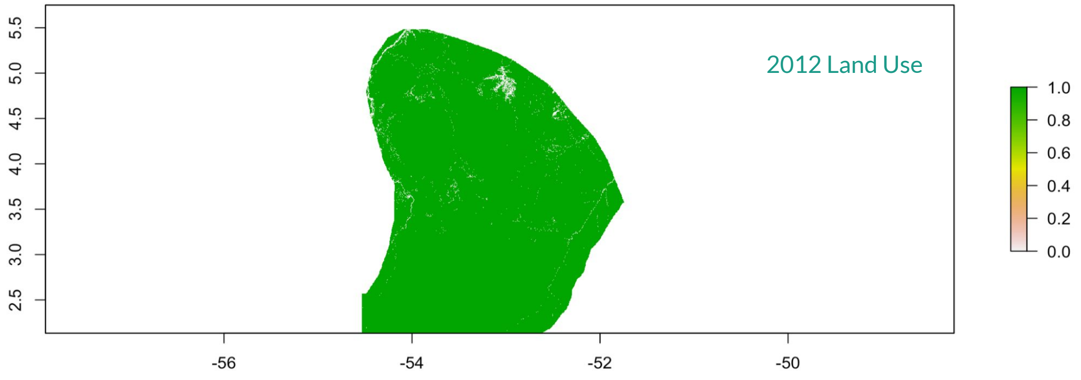
Suitable habitat loss less than 0.2% !

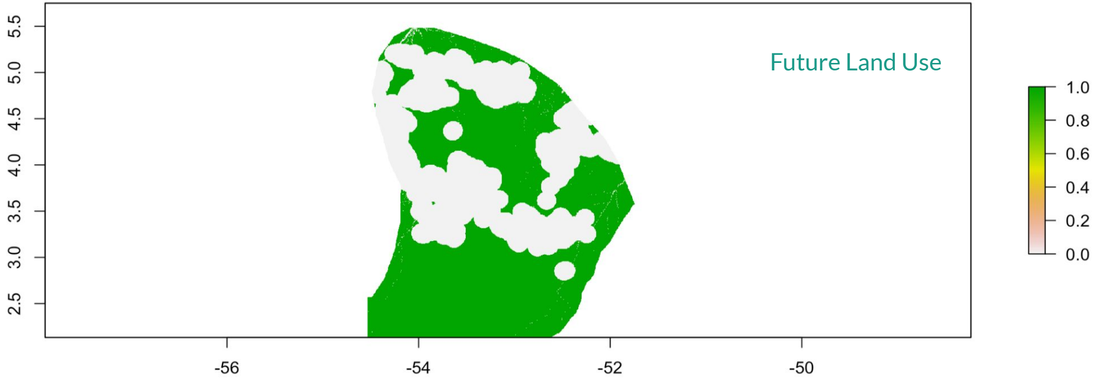
Not in "Threatened Categories", but
Near Threatened or Least Concern?

Future Suitability Projection

```
> ### calculate the suitable habitat in 2012
> a=area(suit_2012_dentin)
> # calculate the favorable area in the future land use
> suitability_fut=reclassify(lu_fut, rcl=rcl)
> suit_fut_dentin=suitability_fut*dentin.r
> plot(suit_fut_dentin)
> a=area(suit_fut_dentin)
> zonal(a, suit_fut_dentin, "sum")
     zone      sum
[1,]    0 34105.73
[2,]    1 50598.93
> (83590.61-50598.93)*100/83590.61
[1] 39.46817
```

39.5% AOO Loss!





39.5% AOO Loss!

A. Population size reduction. Population reduction (measured over the longer of 10 years or 3 generations) based on any of A1 to A4

	Critically Endangered	Endangered	Vulnerable
A1	≥ 90%	≥ 70%	≥ 50%
A2, A3 & A4	≥ 80%	≥ 50%	≥ 30%

A1 Population reduction observed, estimated, inferred, or suspected in the past where the causes of the reduction are clearly reversible AND understood AND have ceased.

A2 Population reduction observed, estimated, inferred, or suspected in the past where the causes of reduction may not have ceased OR may not be understood OR may not be reversible.

A3 Population reduction projected, inferred or suspected to be met in the future (up to a maximum of 100 years) *[(a) cannot be used for A3]*.

A4 An observed, estimated, inferred, projected or suspected population reduction where the time period must include both the past and the future (up to a max. of 100 years in future), and where the causes of reduction may not have ceased OR may not be understood OR may not be reversible.

based on any of the following:

(a) direct observation *[except A3]*

(b) an index of abundance appropriate to the taxon

(c) a decline in area of occupancy (AOO), extent of occurrence (EOO) and/or habitat quality

(d) actual or potential levels of exploitation

(e) effects of introduced taxa, hybridization, pathogens, pollutants, competitors or parasites.

...will be Vulnerable

Dyeing Poison Dart Frog* shall be listed:

Near Threatened (NT)

**Dendrobates Tinctorius*

A taxon is Near Threatened when it has been evaluated against the criteria but does not qualify for Critically Endangered, Endangered or Vulnerable now, **but is close to qualifying for or is likely to qualify for a threatened category in the near future.**

§ IUCN RED LIST CATEGORIES AND CRITERIA Version 3.1: IUCN (2001)

Thank You!

Sentao ZENG (sentao.zeng@studenti.unimi.it)

Environmental Change and Global Sustainability
Università degli Studi di Milano
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